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BSCS 1-1

A short solution for every number question (1-15)

Q1 — (10000110_2)

(128+4+2=) 134

Answer: $134^{10}$

Q2 — (3466_8)

(3(512)+4(64)+6(8)+6(1))

(1536+256+48+6=) $1846^{10}$

Answer: $1846^{8}$

Q3 — (7F5_16)

(7(256)+15(16)+5)

(1792+240+5=) 2037

Answer: $2037^{10}$

Q4 — (113_10) to Binary

113 ÷ 2: 56 R1

56 ÷ 2: 28 R0

28 ÷ 2: 14 R0

14 ÷ 2: 7 R0

7 ÷ 2: 3 R1

3 ÷ 2: 1 R1

1 ÷ 2: 0 R1 : Read remainders bottom → top: $1110001_2$ Final answer: $1110001^{2}$

Edit with WPS Office

Q5 — (2315_10rightarrow octal)

2315÷8 - 289 R3

289÷8 - 36 R1

36÷8 - 4 R4

4÷8  0 R4

Read bottom -  top:

$4413_8$

Q6 — (3290_10rightarrow hexadecimal)

3290÷16 -  205 R10 = A

205÷16 - 12 R13 = D

12÷16 - 0 R12 = C

Bottom  - top:

$CDA_{16}$

Q7 — Binary -  Hex

10111000100

Group: 0101 | 1100 | 0100

= 5 | C | 4

= $5C4_{16}$

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Q8 — Binary - Octal

1011010100

Group: 001 | 011 | 010 | 100

= 1 | 3 | 2 | 4

$=1324_8$

Q9 — Hex → Binary

F = 1111

5 = 0101

B = 1011

$=111101011011_2$

Q10 — Octal → Hex

5 = 101

2 = 010

2 = 010

Binary: 101010010

Group: 0001 | 0101 | 0010

= 1 | 5 | 2

$= 152_{16}$

Q11 - IP 10.18.44.244 to 8-bit:

10 = 00001010

18 = 00010010

44 = 00101100

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244 = 11110100

Q12 - Web colors:

a) #90FFFF = R 144, G 255, B 255

b) rgb(82,53,11) = R 52, G 35, B 0B, Full #52350B

Q13 - chmod 754:

Owner 7 = 111 = rwx

Group 5 = 101 = r-x

Others 4 = 100 = r--

Full = rwxr-xr--

Q14 - 11-bit sensor:

Values = 2048

Largest dec = 2047

Largest bin = $11111111111_2$

Largest hex = $7FF_{16}$

Q15 - My Number:

N = last 3 digits ng ID mo, if less than 100 add 500

Ex N=256

Bin = 1000000002

Oct = 4008

Hex = 10016

Error analysis: Mistake is reading remainders top to bottom. 0111012 is 29 not 46. Correct is 1011102. Always read bottom to top.

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